

National University of Computer & Emerging Sciences

MT2005 – Probability and Statistics

Homework # 1(g)

Measures of Position

You are given a dataset with 6 employee salaries from a company. The data is as follows:

Employee	Salary (Rupees)
1	55000
2	65000
3	70000
4	75000
5	90000
6	150000

1. Z-Scores

- Calculate the **z-scores** for each salary in the dataset.
- Based on your z-scores, identify which salaries could be considered outliers (using a z-score threshold of 3 or -3).

2. Percentiles

- Calculate the **10th** and **90th** percentiles of the salary dataset.
- What do these percentiles tell you about the distribution of salaries?

3. Percentile Rank

- Calculate the **percentile rank** of 75000.
- Interpret your result.

4. Deciles

- Calculate the **4th** and **7th** deciles of the salary dataset.
- What do these deciles tell you about the distribution of salaries?

5. Quartiles

- Calculate the **1st**, **2nd** and **3rd** quartile of the salary data.
- What do these quartiles indicate about the central tendency and spread of salaries?

6. Identifying Outliers

- Use the **Interquartile Range (IQR)** method to detect any outliers in the salary dataset.
- Compare this with the outliers identified using the z-score method. Do they match?

7. Skewness and Kurtosis

- Calculate the **skewness** and **kurtosis** of the salary distribution.
- Based on the skewness value, determine whether the distribution of salaries is positively skewed, negatively skewed, or approximately symmetric.
- Based on the kurtosis value, determine whether the distribution of salaries is mesokurtic, leptokurtic, or platykurtic.

8. Boxplot

- a) Create a **boxplot** of the salary dataset. From the boxplot, visually identify any potential outliers and explain their significance.
- b) Use Python to create the boxplot of the salary dataset.

Python Code for Boxplot:

```
import matplotlib.pyplot as plt
import seaborn as sns

# Student Roll Number (replace with your actual roll number)
roll_number = "Your Roll Number Here"

# Updated salary data
salaries = [55000, 65000, 70000, 75000, 90000, 150000]

# Create a boxplot
plt.figure(figsize=(6, 4))
sns.boxplot(x=salaries)

# Add title with roll number placeholder
plt.title(f"Boxplot of Salaries (Roll Number: {roll_number})")

# Display the boxplot
plt.show()
```